

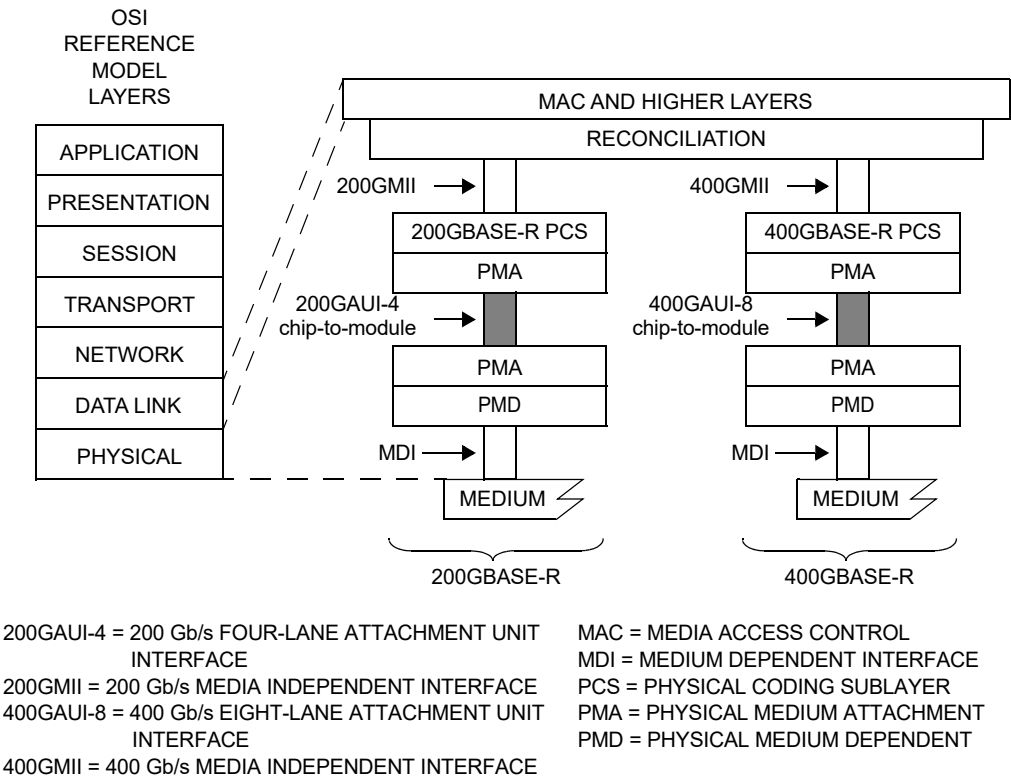
Annex 120E

(normative)

Chip-to-module 200 Gb/s four-lane Attachment Unit Interface (200GAUI-4 C2M) and 400 Gb/s eight-lane Attachment Unit Interface (400GAUI-8 C2M)

120E.1 Overview

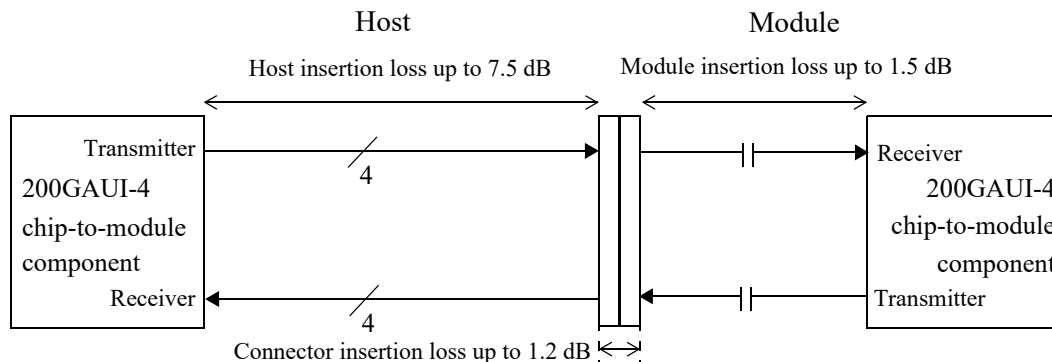
This annex defines the functional and electrical characteristics for the optional chip-to-module 200 Gb/s four-lane Attachment Unit Interface (200GAUI-4 C2M) and 400 Gb/s eight-lane Attachment Unit Interface (400GAUI-8 C2M). Figure 120E–1 shows the relationship of the 200GAUI-4 and 400GAUI-8 C2M interfaces to the ISO/IEC Open System Interconnection (OSI) reference model. The chip-to-module interface provides electrical characteristics and associated compliance points, which can optionally be used when designing systems with pluggable module interfaces.



**Figure 120E–1—Example 200GAUI-4 and 400GAUI-8 chip-to-module relationship to the ISO/IEC Open System Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model**

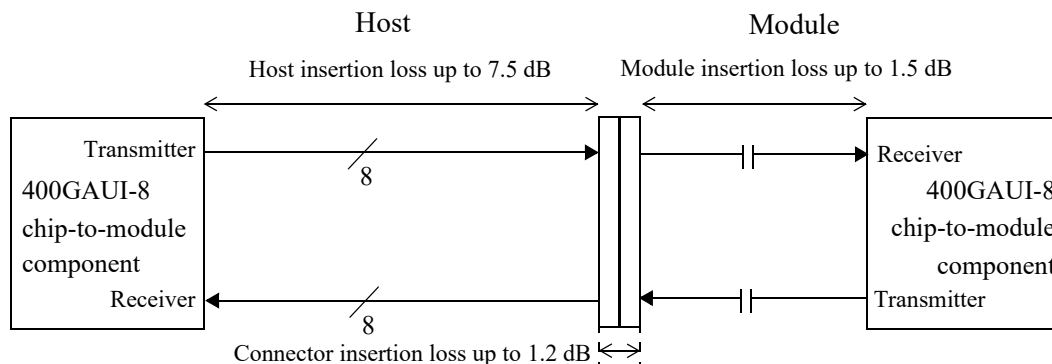
The sublayers (including the PCS and associated FEC) for each PHY that can optionally include a 200GAUI-4 C2M or 400GAUI-8 C2M are summarized in the tables in 116.1.4 and are specified in the corresponding PMD clause. The positioning of the 200GAUI-4 C2M or 400GAUI-8 C2M relative to other sublayers is shown in 120.1 with further examples in Annex 120A.

The 200GAUI-4 C2M link is described in terms of a host 200GAUI-4 C2M component, a 200GAUI-4 C2M channel with associated insertion loss, and a module 200GAUI-4 C2M component. Figure 120E-2 depicts a typical 200GAUI-4 C2M application and summarizes the differential insertion loss budget associated with the chip-to-module application. The supported insertion loss budget is characterized by Equation (120E-1) and illustrated in Figure 120E-4. The 200GAUI-4 C2M interface comprises independent data paths in each direction. Each data path contains four differential lanes using PAM4 signaling, where the highest differential level corresponds to the symbol three and the lowest level corresponds to the symbol zero. Each lane is AC-coupled within the module.



**Figure 120E-2—200GAUI-4 chip-to-module insertion loss budget at 13.28 GHz**

The 400GAUI-8 C2M link is described in terms of a host 400GAUI-8 C2M component, a 400GAUI-8 C2M channel with associated insertion loss, and a module 400GAUI-8 C2M component. Figure 120E-3 depicts a typical 400GAUI-8 C2M application and summarizes the differential insertion loss budget associated with the chip-to-module application. The recommended insertion loss budget is characterized by Equation (120E-1) and illustrated in Figure 120E-4. The 400GAUI-8 C2M interface comprises independent data paths in each direction. Each data path contains eight differential lanes using PAM4 signaling, where the highest differential level corresponds to the symbol three and the lowest level corresponds to the symbol zero. Each lane is AC-coupled within the module.



**Figure 120E-3—400GAUI-8 chip-to-module insertion loss budget at 13.28 GHz**

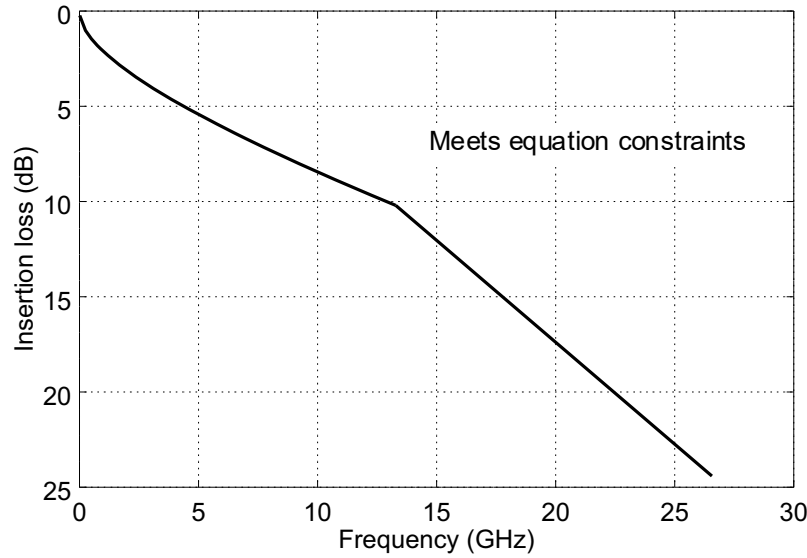
The nominal signaling rate for each lane is 26.5625 GBd. The chip-to-module interface is defined using a specification and test methodology that is similar to that used for CEI-56G-VSR-PAM defined in OIF-CEI-04.0 [B58].

$$Insertion\_loss(f) \leq \begin{cases} 0.05 + 1.8\sqrt{f} + 0.2705f & 0.01 \leq f \leq 13.28 \\ -4.0096 + 1.07f & 13.28 < f \leq 26.5625 \end{cases} \text{ (dB)} \quad (120E-1)$$

where

$f$  is the frequency in GHz

$Insertion\_loss(f)$  is the 200GAUI-4 or 400GAUI-8 chip-to-module insertion loss



**Figure 120E-4—Recommended 200GAUI-4 or 400GAUI-8 chip-to-module channel insertion loss**

### 120E.1.1 Bit error ratio

The bit error ratio (BER) when processed according to Clause 120 shall be less than  $10^{-5}$ .

### 120E.2 200GAUI-4 and 400GAUI-8 chip-to-module compliance point definitions

The electrical characteristics for the 200GAUI-4 and 400GAUI-8 chip-to-module interfaces are defined at compliance points for the host and module, respectively. Reference test fixtures, called compliance boards, are used to access the electrical specification parameters. Figure 120E-5 depicts the location of compliance points when measuring host 200GAUI-4 or 400GAUI-8 C2M compliance. The output of the Host Compliance Board (HCB) is used to verify the host electrical output signal at TP1a. Similarly, the input of the HCB at TP4a is used to verify the host input compliance.

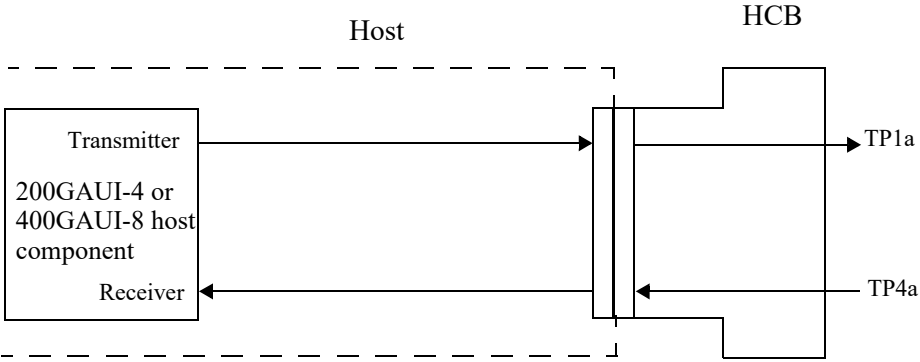


Figure 120E-5—Host 200GAUI-4 or 400GAUI-8 C2M compliance points

Figure 120E-6 depicts the location of compliance points when measuring module 200GAUI-4 or 400GAUI-8 C2M compliance. The output of the Module Compliance Board (MCB) is used to verify the module electrical output signal at TP4. Similarly, the input of the MCB at TP1 is used to verify the module input compliance. Additional details on the requirements for the MCB and HCB are given in 120E.4.1.

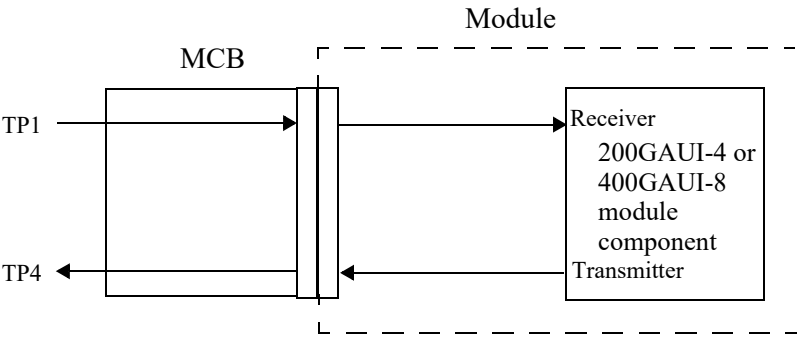


Figure 120E-6—Module 200GAUI-4 or 400GAUI-8 C2M compliance points

120E.3 200GAUI-4 and 400GAUI-8 chip-to-module electrical characteristics

120E.3.1 200GAUI-4 and 400GAUI-8 C2M host output characteristics

A 200GAUI-4 or a 400GAUI-8 C2M host output shall meet the specifications given in Table 120E-1.

A test system with a fourth-order Bessel-Thomson low-pass response with 33 GHz 3 dB bandwidth is to be used for all output signal measurements, unless otherwise specified.

**Table 120E-1—200GAUI-4 and 400GAUI-8 C2M host output characteristics at TP1a**

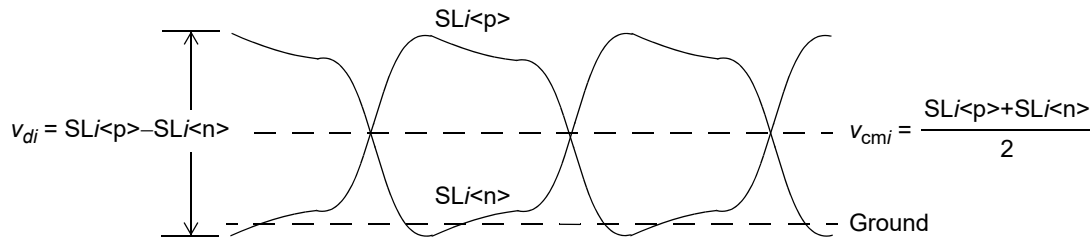
| Parameter                                                                                     | Reference  | Value             | Units |
|-----------------------------------------------------------------------------------------------|------------|-------------------|-------|
| Signaling rate per lane (range)                                                               | 120E.3.1.1 | 26.5625 ± 100 ppm | GBd   |
| DC common-mode output voltage (max)                                                           | 120E.3.1.2 | 2.8               | V     |
| DC common-mode output voltage (min)                                                           | 120E.3.1.2 | −0.3              | V     |
| Single-ended output voltage (max)                                                             | 120E.3.1.2 | 3.3               | V     |
| Single-ended output voltage (min)                                                             | 120E.3.1.2 | −0.4              | V     |
| AC common-mode output voltage (max, RMS)                                                      | 120E.3.1.2 | 17.5              | mV    |
| Differential peak-to-peak output voltage (max)<br>Transmitter disabled<br>Transmitter enabled | 120E.3.1.2 | 35<br>880         | mV    |
| ESMW (Eye symmetry mask width)                                                                | 120E.4.2   | 0.22              | UI    |
| Eye height, differential (min)                                                                | 120E.4.2   | 32                | mV    |
| Vertical eye closure (max)                                                                    | 120E.4.3   | 12                | dB    |
| Differential output return loss (min)                                                         | 83E.3.1.3  | Equation (83E-2)  | dB    |
| Common to differential mode conversion return loss (min)                                      | 83E.3.1.3  | Equation (83E-3)  | dB    |
| Differential termination mismatch (max)                                                       | 120E.3.1.4 | 10                | %     |
| Transition time (min, 20% to 80%)                                                             | 120E.3.1.5 | 10                | ps    |

### 120E.3.1.1 Signaling rate and range

The 200GAUI-4 and 400GAUI-8 C2M signaling rate is 26.5625 GBd ± 100 ppm per lane. This translates to a nominal unit interval of 37.647059 ps.

### 120E.3.1.2 Signal levels

The differential output voltage  $v_{di}$  is defined to be the difference between the single-ended output voltages,  $SL_{i<p>}$  minus  $SL_{i<n>}$ . The common-mode voltage  $v_{cmi}$  is defined to be one half of the sum of  $SL_{i<p>}$  and  $SL_{i<n>}$ . These definitions are illustrated by Figure 120E-7.



**Figure 120E-7—Voltage definitions**

The DC common-mode output voltage and AC common-mode output voltage are defined with respect to signal ground.

Unless otherwise noted, differential and common-mode signal voltages are measured with a PRBS13Q test pattern.

#### **120E.3.1.3 Output return loss**

Differential output return loss is defined in 83E.3.1.3.

#### **120E.3.1.4 Differential termination mismatch**

Differential termination mismatch is defined in 86A.5.3.2.

#### **120E.3.1.5 Transition time**

In this annex, transition times are specified for transitions between three consecutive “zero” symbols and three consecutive “three” symbols, or vice versa. The specified times are between the crossings of 20% and 80% levels of the signal.

The test pattern used is PRBS13Q, the transitions within sequences of three zeros and three threes, and three threes and three zeros, respectively, are measured. These are PAM4 symbols 1820 to 1825 and 2086 to 2091, respectively, where symbols 1 to 7 are the run of seven threes. The 0% level and the 100% level are defined as the average signal within windows from  $-1.5$  UI to  $-1$  UI and from  $1.5$  UI to  $2$  UI relative to the edge.

The waveform is observed through a fourth-order Bessel-Thomson low-pass response with a 3 dB bandwidth of 33 GHz.

NOTE—This definition is not the same as the rise and fall times typically reported by an oscilloscope from an eye diagram, which take all the edges into account.

#### **120E.3.1.6 Host output eye width and eye height**

Figure 120E–8 depicts an example host output eye width and eye height test configuration. Host output eye width and eye height are measured at TP1a using compliance boards defined in 120E.4.1. The host output eye is measured using a reference receiver with a continuous time linear equalizer (CTLE) defined in 120E.3.1.7. Eye width and eye height measurement methodology is described in 120E.4.2. All counter-propagating signals shall be asynchronous to the co-propagating signals using the PRBS13Q or PRBS31Q pattern, or a valid 200GBASE-R or 400GBASE-R signal. PRBS13Q is described in 120.5.11.2.1. PRBS31Q is described in 120.5.11.2.2. For the case where PRBS13Q or PRBS31Q are used with a common clock, there is at least 31 UI delay between the patterns on one lane and any other lane, so that the symbols on each lane are not correlated. The crosstalk generator is calibrated at TP4 (without the use of a CTLE) with target differential peak-to-peak amplitude of 900 mV and slew time of 12 ps between  $\pm 0.27$  V.

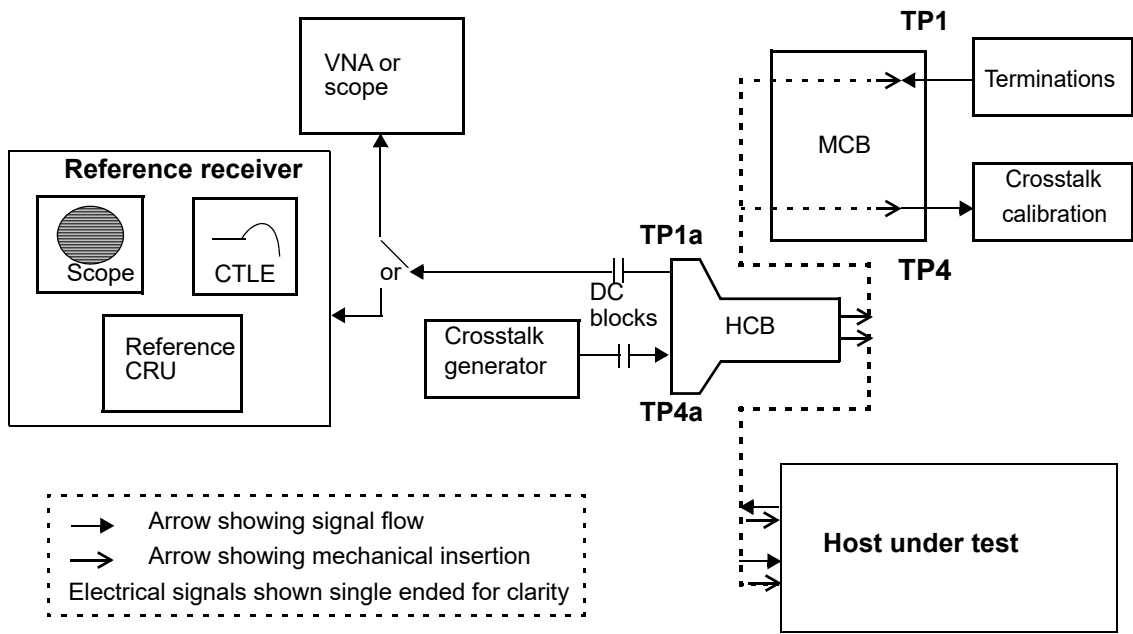


Figure 120E-8—Example host output test configuration

120E.3.1.7 Reference receiver for eye width and eye height evaluation

The reference receiver is used to measure eye width and eye height. The reference receiver includes a selectable continuous time linear equalizer (CTLE), which is described by Equation (120E-2) with coefficients given in Table 120E-2 and illustrated in Figure 120E-9. The equalizer may be implemented in software; however, the measured signal is not averaged.

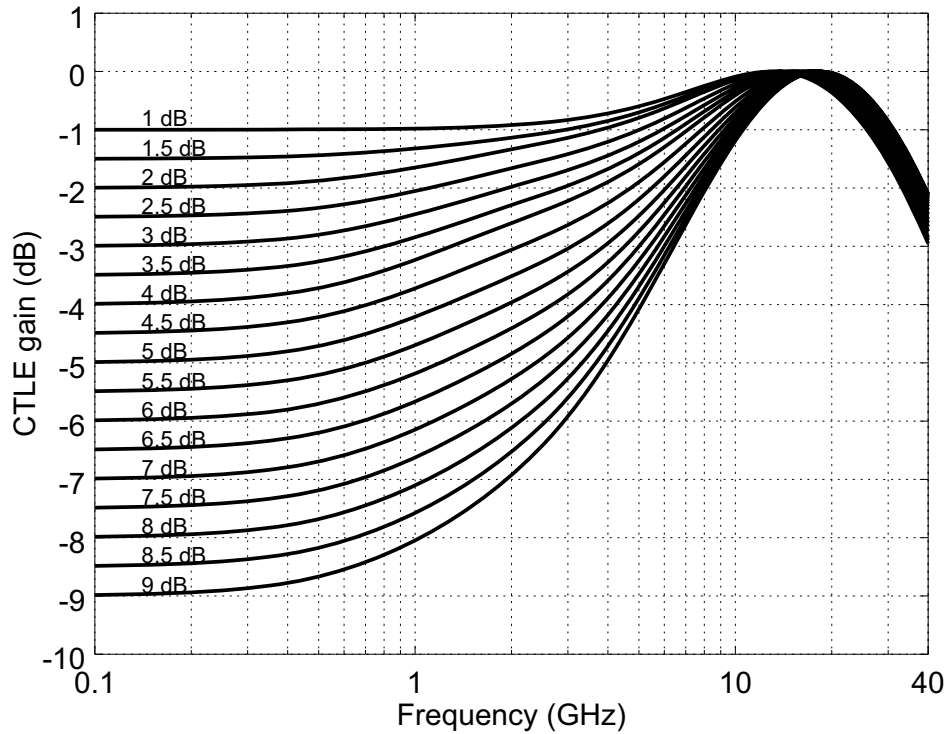
$$H(f) = \frac{GP_1P_2P_{LF}}{Z_1Z_{LF}} \times \frac{j2\pi f + Z_1}{(j2\pi f + P_1)(j2\pi f + P_2)} \times \frac{j2\pi f + Z_{LF}}{j2\pi f + P_{LF}} \quad (120E-2)$$

where

|            |                                          |
|------------|------------------------------------------|
| $H(f)$     | is the CTLE transfer function            |
| $G$        | is the CTLE gain                         |
| $P_1, P_2$ | are the CTLE poles in Grad/s             |
| $Z_1$      | is the CTLE zero in Grad/s               |
| $P_{LF}$   | is the low frequency CTLE pole in Grad/s |
| $Z_{LF}$   | is the low frequency CTLE zero in Grad/s |
| $j$        | is the square root of $-1$               |
| $f$        | is the frequency in GHz                  |

**Table 120E-2—Reference CTLE coefficients**

| Peaking (dB) | G        | $\frac{P_1}{2\pi}$ | $\frac{P_2}{2\pi}$ | $\frac{Z_1}{2\pi}$ | $\frac{P_{LF}}{2\pi}$ | $\frac{Z_{LF}}{2\pi}$ |
|--------------|----------|--------------------|--------------------|--------------------|-----------------------|-----------------------|
| 1            | 0.891251 | 26.5625            | 14.1               | 9.463748           | 1.2                   | 1.2                   |
| 1.5          | 0.841395 | 26.5625            | 14.1               | 9.248465           | 1.2                   | 1.15                  |
| 2            | 0.794328 | 26.5625            | 14.1               | 9.069645           | 1.2                   | 1.1                   |
| 2.5          | 0.749894 | 26.5625            | 14.1               | 8.640319           | 1.2                   | 1.075                 |
| 3            | 0.707946 | 26.5625            | 14.1               | 8.255665           | 1.2                   | 1.05                  |
| 3.5          | 0.668344 | 26.5625            | 14.1               | 7.906766           | 1.2                   | 1.025                 |
| 4            | 0.630957 | 26.5625            | 14.1               | 7.58765            | 1.2                   | 1                     |
| 4.5          | 0.595662 | 26.5625            | 14.1               | 7.076858           | 1.2                   | 1                     |
| 5            | 0.562341 | 26.5625            | 14.1               | 6.614781           | 1.2                   | 1                     |
| 5.5          | 0.530884 | 26.5625            | 14.1               | 6.193091           | 1.2                   | 1                     |
| 6            | 0.501187 | 26.5625            | 14.1               | 5.805801           | 1.2                   | 1                     |
| 6.5          | 0.473151 | 26.5625            | 14.1               | 5.448395           | 1.2                   | 1                     |
| 7            | 0.446684 | 26.5625            | 14.1               | 5.117337           | 1.2                   | 1                     |
| 7.5          | 0.421697 | 26.5625            | 14.1               | 4.809777           | 1.2                   | 1                     |
| 8            | 0.398107 | 26.5625            | 14.1               | 4.523367           | 1.2                   | 1                     |
| 8.5          | 0.375837 | 26.5625            | 14.1               | 4.256129           | 1.2                   | 1                     |
| 9            | 0.354813 | 26.5625            | 14.1               | 4.006377           | 1.2                   | 1                     |



**Figure 120E-9—Reference continuous time linear equalizer (CTLE) characteristic**

### 120E.3.2 200GAUI-4 and 400GAUI-8 C2M module output characteristics

A 200GAUI-4 or a 400GAUI-8 C2M module output shall meet the specifications given in Table 120E-3. The low-frequency 3 dB cutoff of the output AC-coupling within the module should be less than 100 kHz. A test system with a fourth-order Bessel-Thomson low-pass response with 33 GHz 3 dB bandwidth is to be used for all output signal measurements, unless otherwise specified.

**Table 120E-3—200GAUI-4 and 400GAUI-8 C2M module output characteristics at TP4**

| Parameter                                      | Reference    | Value                 | Units |
|------------------------------------------------|--------------|-----------------------|-------|
| Signaling rate per lane (range)                | 120E.3.1.1   | $26.5625 \pm 100$ ppm | GBd   |
| AC common-mode output voltage (max, RMS)       | 120E.3.1.2   | 17.5                  | mV    |
| Differential peak-to-peak output voltage (max) | 120E.3.1.2   | 900                   | mV    |
| Near-end ESMW (Eye symmetry mask width)        | 120E.4.2     | 0.265                 | UI    |
| Near-end Eye height, differential (min)        | 120E.4.2     | 70                    | mV    |
| Far-end ESMW (Eye symmetry mask width)         | 120E.4.2     | 0.2                   | UI    |
| Far-end Eye height, differential (min)         | 120E.4.2     | 30                    | mV    |
| Far-end pre-cursor ISI ratio                   | 120E.3.2.1.2 | -4.5 to +2.5          | %     |

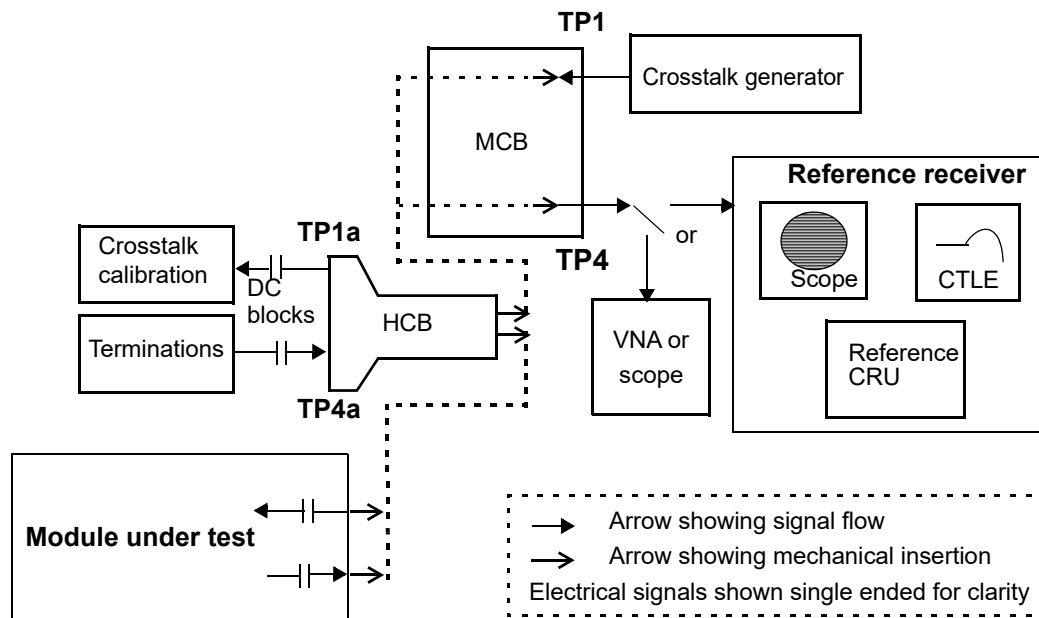
**Table 120E–3—200GAUI-4 and 400GAUI-8 C2M module  
output characteristics at TP4 (*continued*)**

| Parameter                                            | Reference  | Value            | Units |
|------------------------------------------------------|------------|------------------|-------|
| Differential output return loss (min)                | 83E.3.1.3  | Equation (83E–2) | dB    |
| Common-to-differential-mode output return loss (min) | 83E.3.1.3  | Equation (83E–3) | dB    |
| Differential termination mismatch (max)              | 120E.3.1.4 | 10               | %     |
| Transition time (min, 20% to 80%)                    | 120E.3.1.5 | 9.5              | ps    |
| DC common-mode voltage (min) <sup>a</sup>            | 120E.3.1.2 | –350             | mV    |
| DC common-mode voltage (max) <sup>a</sup>            | 120E.3.1.2 | 2850             | mV    |

<sup>a</sup> DC common-mode voltage is generated by the host. Specification includes effects of ground offset voltage.

### 120E.3.2.1 Module output eye width, eye height, and pre-cursor ISI ratio

Figure 120E–10 depicts an example module output eye width and eye height test configuration. Module output eye width and eye height are measured at TP4 using compliance boards defined in 120E.4.1. The module output eye is measured using a reference receiver with a continuous time linear equalizer (CTLE) defined in 120E.3.2.1.1. Eye width and eye height measurement methodology is described in 120E.4.2. All counter-propagating signals shall be asynchronous to the co-propagating signals using the PRBS13Q or PRBS31Q pattern, or a valid 200GBASE-R or 400GBASE-R signal. PRBS13Q is described in 120.5.11.2.1 and PRBS31Q is described in 120.5.11.2.2. For the case where PRBS13Q or PRBS31Q are used with a common clock, there is at least 31 UI delay between the patterns on one lane and any other lane, so that the symbols on each lane are not correlated. The crosstalk generator is calibrated at TP1a (without the use of a CTLE) with target differential peak-to-peak amplitude of 880 mV and target transition time of 19 ps.



**Figure 120E–10—Example module output test configuration**

### 120E.3.2.1.1 Reference receiver for module output evaluation

A reference receiver is used to measure module eye width and eye height. The reference receiver includes a selectable continuous time linear equalizer (CTLE), which is defined in 120E.3.1.7. The equalizer may be implemented in software; however, the measured signal is not averaged.

The near-end eye is measured with the reference receiver. Equalizer settings corresponding to less than or equal to 3 dB of peaking from Table 120E-2 may be used to meet the output eye width and height requirements.

For the far-end eye, the signal measured at TP4 is first convolved with a loss channel (~6.4 dB loss at Nyquist) that represents the worst case channel loss. The loss channel is the host trace defined in 92.10.7.1.1 with  $Z_p = 151$  mm. The reference receiver is then used to measure the eye width and height. Any of the equalizer settings from Table 120E-2 may be used.

### 120E.3.2.1.2 Far-end pre-cursor ISI ratio

Capture the PRBS13Q waveform corresponding to the far-end eye (see 120E.3.2.1.1) and calculate the linear fit pulse using the procedure defined in 120D.3.1.3. Any setting of the reference CTLE for which the far-end eye width and height satisfy the limits in Table 120E-3, may be used.

The peak amplitude of the linear fit pulse is  $p_{\max}$ . The pre-cursor ISI  $p_{\text{pre}}$  is the value of the linear fit pulse 1 UI prior to the time of the pulse peak. The pre-cursor ISI ratio is  $p_{\text{pre}} / p_{\max}$ .

### 120E.3.3 200GAUI-4 and 400GAUI-8 C2M host input characteristics

A 200GAUI-4 or a 400GAUI-8 C2M host input shall meet the specifications given in Table 120E-4.

**Table 120E-4—200GAUI-4 and 400GAUI-8 C2M host input characteristics**

| Parameter                                           | Reference  | Test point | Value                 | Units |
|-----------------------------------------------------|------------|------------|-----------------------|-------|
| Signaling rate, per lane (range)                    | 120E.3.1.1 | TP4a       | $26.5625 \pm 100$ ppm | GBd   |
| Differential pk-pk input voltage tolerance (min)    | 120E.3.1.2 | TP4        | 900                   | mV    |
| Differential input return loss (min)                | 83E.3.3.1  | TP4a       | Equation (83E-5)      | dB    |
| Differential-to-common-mode input return loss (min) | 83E.3.3.1  | TP4a       | Equation (83E-6)      | dB    |
| Host stressed input test <sup>a</sup>               | 120E.3.3.2 | TP4        | See 120E.3.3.2        |       |
| Differential termination mismatch (max)             | 120E.3.1.4 | TP4a       | 10                    | %     |
| Common-mode voltage <sup>b</sup><br>Min<br>Max      | 120E.3.1.2 | TP4a       | -0.3<br>2.8           | V     |

<sup>a</sup>Meets BER specified in 120E.1.1.

<sup>b</sup>Generated by host, referred to host ground.

### 120E.3.3.1 Input return loss

Differential input return loss is defined in 83E.3.3.1.

### 120E.3.3.2 Host stressed input test

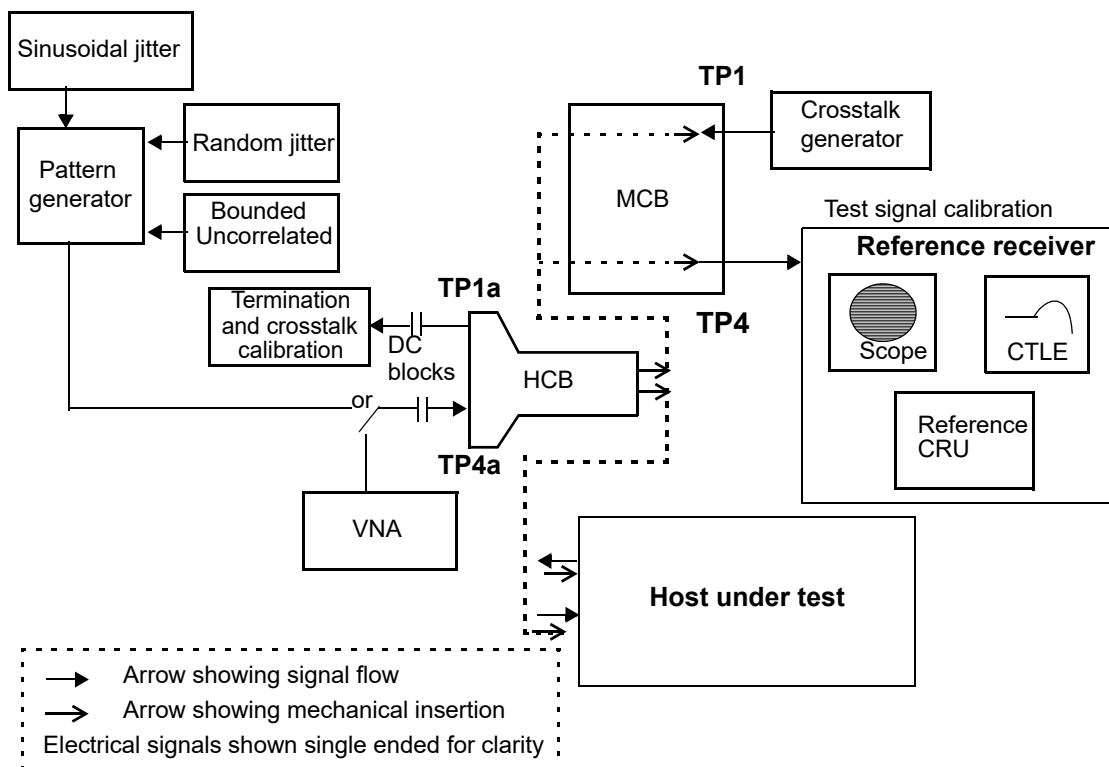
The host stressed input tolerance is measured using the procedure defined in 120E.3.3.2.1. The input shall satisfy the input tolerance with the parameters in Table 120E–5.

**Table 120E–5—Host stressed input parameters**

| Parameter                                 | Value        |
|-------------------------------------------|--------------|
| Far-end ESMW<br>(Eye symmetry mask width) | 0.2 UI       |
| Far-end Eye width                         | 0.2 UI       |
| Applied pk-pk sinusoidal jitter           | Table 120E–6 |
| Far-end Eye height                        | 30 mV        |

#### 120E.3.3.2.1 Host stressed input test procedure

The host stressed input test is summarized in Figure 120E–11. The stressed signal is applied at TP4a and is calibrated at TP4. A reference CRU with a corner frequency of 4 MHz and slope of 20 dB/decade is used to calibrate the stressed signal using the PRBS13Q Pattern (see 120.5.11.2.1). The reference receiver includes a selectable CTLE as specified in 120E.3.2.1.1. The stressed signal is generated by adding sinusoidal jitter, random jitter, and bounded uncorrelated jitter to a clean signal. The frequency and peak-to-peak amplitude of the sinusoidal jitter used for the host stressed input test is given in Table 120E–6.



**Figure 120E–11—Example host stressed input test**

**Table 120E-6—Applied sinusoidal jitter**

| Parameter                | Case A | Case B | Case C | Case D | Case E | Case F | Units |
|--------------------------|--------|--------|--------|--------|--------|--------|-------|
| Jitter frequency         | 0.04   | 0.4    | 1.333  | 4      | 12     | 40     | MHz   |
| Jitter amplitude (pk-pk) | 5      | 0.5    | 0.15   | 0.05   | 0.05   | 0.05   | UI    |

Bounded uncorrelated jitter provides a source of bounded high probability jitter uncorrelated with the pattern. This jitter stress source may not be present in all stressed pattern generators or bit error ratio testers. It can be generated by driving the pattern generator external jitter modulation input with a filtered PRBS pattern. The PRBS pattern length should be between PRBS7 and PRBS9 with a signaling rate approximately 1/10 of the stressed pattern signaling rate (e.g., 2.65625 Gb/s). The clock source for the PRBS generator is asynchronous to the pattern generator clock source to assure non-correlation of the jitter. The low-pass filter that operates on the PRBS pattern to generate the bounded uncorrelated jitter should exhibit 20 dB/decade roll-off with a  $-3$  dB corner frequency between 150 MHz and 300 MHz. This value also has to be below the upper frequency limit of the pattern generator external modulator input. Random jitter and bounded uncorrelated jitter are added such that the output of the pattern generator approximates the 200GAUI-4 and 400GAUI-8 C2C output jitter profile given in Table 120D-1.

The counter propagating crosstalk signals during calibration of the stressed signal are asynchronous with target amplitude of 880 mV peak-to-peak differential and 20% to 80% target transition time of 19 ps as measured at TP1a (without the use of a CTLE). The crosstalk signal transition time is calibrated with PRBS13Q. The pattern may be changed to a valid 400GBASE-R signal for amplitude calibration and the stressed input test. The PRBS13Q pattern is described in 120.5.11.2.1. For the case where the PRBS13Q pattern is used with a common clock, there is at least 31 UI delay between the PRBS13Q patterns on one lane and any other lane, so that the symbols on each lane are not correlated. Any one of these patterns is sufficient as a crosstalk aggressor with all lanes active during the stressed input test.

The far-end eye height and eye width, measured to a probability of  $10^{-5}$ , are then measured at TP4 with the reference receiver defined in 120E.3.2.1.1 using the measurement methodology given in 120E.4.2. Note that the reference receiver for far-end eye height and width measurements includes a loss channel. Random jitter and the pattern generator output levels are adjusted (without exceeding the differential pk-pk input voltage tolerance specification as shown in Table 120E-4) to result in the eye height for all three eyes and eye width for the smallest eye given in Table 120E-5 with the setting of the CTLE that maximizes the product of eye height and eye width.

The far-end pre-cursor ISI ratio is measured using the method defined in 120E.3.2.1.2 and it shall meet the specification in Table 120E-3. Pre-emphasis capability is likely to be required in the pattern generator to meet this requirement.

The pattern is then changed to Pattern 5, Pattern 3, or a valid 200GBASE-R/400GBASE-R signal for the input test which is conducted by inserting the HCB into the host under test. Patterns 3 and 5 are described in Table 124-9.

If the test is performed with pattern 3, the host BER may be calculated using the host's PMA test pattern checker (see 120.5.11.2.2). If the test is performed with pattern 5 or a valid 200GBASE-R/400GBASE-R signal, the BER may be calculated using the host's PCS Reed-Solomon decoder error counters (see 119.3.1), as the number of FEC symbol errors divided by the number of received bits. The number of received bits may be estimated based on the test time.

### 120E.3.4 200GAUI-4 and 400GAUI-8 C2M module input characteristics

A 200GAUI-4 or a 400GAUI-8 C2M module input shall meet the specifications given in Table 120E-7. Channel equalization is provided by an adaptive equalizer in the module.

**Table 120E-7—200GAUI-4 and 400GAUI-8 C2M module input characteristics**

| Parameter                                                     | Reference  | Test point | Value                 | Units |
|---------------------------------------------------------------|------------|------------|-----------------------|-------|
| Signaling rate per lane (range)                               | 120E.3.1.1 | TP1        | $26.5625 \pm 100$ ppm | GBd   |
| Differential pk-pk input voltage tolerance (min) <sup>a</sup> | 120E.3.1.2 | TP1a       | 900                   | mV    |
| Differential input return loss (min)                          | 83E.3.3.1  | TP1        | Equation (83E-5)      | dB    |
| Differential-to-common-mode input return loss (min)           | 83E.3.3.1  | TP1        | Equation (83E-6)      | dB    |
| Differential termination mismatch (max)                       | 120E.3.1.4 | TP1        | 10                    | %     |
| Module stressed input test <sup>b</sup>                       | 120E.3.4.1 | TP1a       | See 120E.3.4.1        |       |
| Single-ended voltage tolerance range (min)                    | 120E.3.1.2 | TP1a       | −0.4 to 3.3           | V     |
| DC common-mode voltage (min) <sup>c</sup>                     | 120E.3.1.2 | TP1        | −350                  | mV    |
| DC common-mode voltage (max) <sup>c</sup>                     | 120E.3.1.2 | TP1        | 2850                  | mV    |

<sup>a</sup> With the exception to 120E.3.1.2 that the pattern is PRBS31Q or scrambled idle.

<sup>b</sup> Meets BER specified in 120E.1.1.

<sup>c</sup> DC common-mode voltage generated by the host. Specification includes effects of ground offset voltage.

#### 120E.3.4.1 Module stressed input test

The module stressed input tolerance is measured using the procedure defined in 120E.3.4.1.1. The input shall satisfy the input tolerance with the parameters in Table 120E-8.

**Table 120E-8—Module stressed input parameters**

| Parameter                       | Value        |
|---------------------------------|--------------|
| ESMW (Eye symmetry mask width)  | 0.22 UI      |
| Eye width                       | 0.22 UI      |
| Applied pk-pk sinusoidal jitter | Table 120E-6 |
| Eye height                      | 32 mV        |

##### 120E.3.4.1.1 Module stressed input test procedure

The module stressed input test is summarized in Figure 120E-12. The stressed signal is applied at TP1 and is calibrated at TP1a. A reference CRU with a corner frequency of 4 MHz and slope of 20 dB/decade is used to calibrate the stressed signal using a PRBS13Q pattern. The reference receiver includes a selectable CTLE as specified in 120E.3.1.7. The stressed signal is generated by adding sinusoidal jitter, random jitter, and

bounded uncorrelated jitter to a clean signal, followed by frequency-dependent attenuation. The frequency-dependent attenuator represents the host channel and may be implemented with PCB traces. The amount of applied peak-to-peak sinusoidal jitter used for the module stressed input test is given in Table 120E–6. Bounded uncorrelated jitter provides a source of bounded high probability jitter uncorrelated with the signal stream. This jitter stress source may not be present in all stressed pattern generators or bit error ratio testers. It can be generated by driving the pattern generator external jitter modulation input with a filtered PRBS pattern. The PRBS pattern length should be between PRBS7 and PRBS9 with a signaling rate approximately 1/10 of the stressed pattern signaling rate (e.g., 2.65625 Gb/s). The clock source for the PRBS generator is asynchronous to the pattern generator clock source to assure non-correlation of the jitter. The low-pass filter that operates on the PRBS pattern to generate the bounded uncorrelated jitter should exhibit 20 dB/decade roll-off with a –3 dB corner frequency between 150 MHz and 300 MHz. This value also has to be below the upper frequency limit of the pattern generator external modulator input. Random jitter and bounded uncorrelated jitter are added such that the output of the pattern generator approximates the 200GAUI-4 and 400GAUI-8 C2C output jitter profile given in Table 120D–1. The target pattern generator 20% to 80% transition time at the input to the test channel in the module stressed input test is 10.5 ps. The return loss of the test system as measured at TP1 meets the specification given in Equation (83E–2).

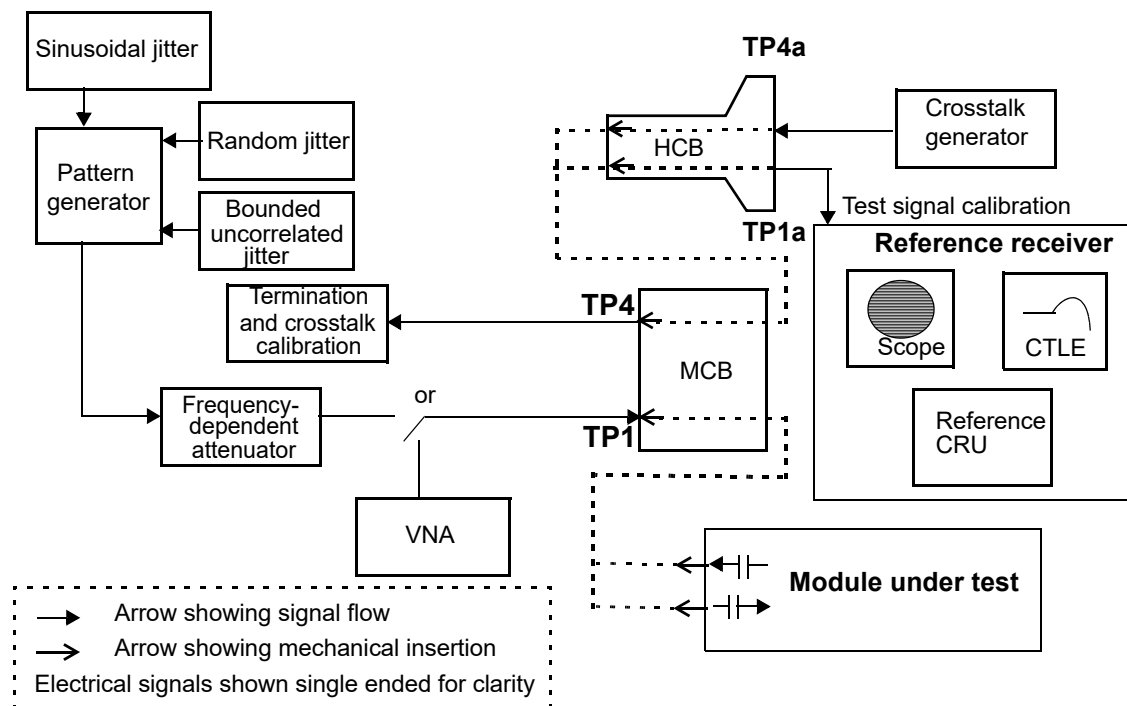


Figure 120E–12—Example module stressed input test

For the high loss case, pre-emphasis capability is likely to be required in the pattern generator to meet the TP1a eye height and eye width specifications.

The counter propagating crosstalk signals during calibration of the stressed signal are asynchronous with target amplitude of 900 mV peak-to-peak differential and target slew time between  $\pm 270$  mV of 12 ps as measured at TP4 (without the use of a CTLE). The crosstalk signal transition time is calibrated with a PRBS13Q pattern. The pattern may be changed to a valid 200GBASE-R/400GBASE-R signal for amplitude calibration and the stressed input test. The PRBS13Q pattern is described in 120.5.11.2.1. For the case where the PRBS13Q pattern is used with a common clock, there is at least 31 UI delay between the PRBS13Q patterns on one lane and any other lane, so that the symbols on each lane are not correlated. Any one of these patterns is sufficient as a crosstalk aggressor with all lanes being active during the stressed input test.

Two levels of frequency-dependent attenuation are used for the module stressed input test: high loss and low loss. For the high loss case, frequency-dependent attenuation is added such that the loss at 13.28 GHz from the output of the pattern generator to TP1a is 12.2 dB. The 12.2 dB loss represents 10.5 dB channel loss with an additional allowance for host transmitter package loss. Eye height and eye width are then measured at TP1a based on the eye measurement methodology given in 120E.4.2 and vertical eye closure is measured according to 120E.4.3. Random jitter and the pattern generator output levels are adjusted (without exceeding the differential pk-pk input voltage tolerance specification as shown in Table 120E-7) to result in the eye height for all three eyes and eye width for the smallest eye given in Table 120E-8 using the reference receiver with the setting of the CTLE that maximizes the product of eye height and eye width. This CTLE setting has to be greater than or equal to 7 dB. For the low loss case, discrete frequency-dependent attenuation is removed such that from the output of the pattern generator to TP1a comprises the mated HCB/MCB pair as described in 120E.4.1. Eye height and eye width at TP1a are then adjusted in the same way as described for the high loss case except that the restriction that the CTLE setting has to be greater than or equal to 7 dB does not apply. In both cases, the input vertical eye closure is less than 12 dB.

The pattern is then changed to Pattern 5, Pattern 3, or a valid 200GBASE-R/400GBASE-R signal for the input test, which is conducted by inserting the module into the MCB. Patterns 3 and 5 are described in Table 124-9.

If the test is performed with pattern 3, the module bit errors may be counted using a PMA test pattern checker (see 120.5.11.2.2) if this option is implemented in the module.

If the test is performed with pattern 5 or a valid 200GBASE-R/400GBASE-R signal, the module BER may be calculated by placing the module under test into local loopback (see 120.5.9) and feeding the module output into a compliant host or its equivalent. The BER may then be calculated using the host's PCS Reed-Solomon decoder error counters (see 119.3.1), as the number of FEC symbol errors divided by the number of received bits.

Methods of extracting the received bit pattern and counting errors other than the ones described above may be used if they generate equivalent results.

The number of received bits may be estimated based on the test time.

The module 200GAUI-4 or 400GAUI-8 C2M receiver under test shall meet the BER requirement as described in 120E.1.1 for both the high loss test and low loss test.

## **120E.4 200GAUI-4 and 400GAUI-8 C2M measurement methodology**

This subclause describes common measurement tools and methodologies to be used for the 200GAUI-4 and 400GAUI-8 chip-to-module interface. Details of HCB and MCB characteristics are given in 120E.4.1 and details of the eye diagram measurement methodology are given in 120E.4.2.

### **120E.4.1 HCB/MCB characteristics**

HCB characteristics are described in 92.11.1 where the HCB performs the equivalent function as the TP2 or TP3 test fixture with the exception that the upper frequency of 25 GHz is replaced with 26.5625 GHz. The MCB characteristics are described in 92.11.2 where the MCB performs the equivalent functionality as the cable assembly test fixture with the exception that the upper frequency of 25 GHz is replaced with 26.5625 GHz.

The mated compliance board characteristics are described in 92.11.3 where the MCB and HCB perform the equivalent functionality as the cable assembly test fixtures with the exceptions that the upper frequency of

25 GHz is replaced with 26.5625 GHz, MDNEXT shall be less than 1.5 mV RMS, MDFEXT shall be less than 4.2 mV RMS, ICN shall be less than 4.4 mV RMS, FOM<sub>ILD</sub> limit is 0.1 dB, and the reference insertion loss is given in Equation (120E-3) for  $0.01 \text{ GHz} \leq f \leq 25 \text{ GHz}$ .

$$IL(f) = 0.471\sqrt{f} + 0.1194f + 0.002f^2 \quad (\text{dB}) \quad (120E-3)$$

where

$f$  is the frequency in GHz  
 $IL(f)$  is the mated compliance board reference insertion loss

#### 120E.4.2 Eye width and eye height measurement method

Eye diagrams in 200GAUI-4 and 400GAUI-8 chip-to-module are measured using a reference receiver. The reference receiver includes a fourth-order Bessel-Thomson low-pass filter response with 33 GHz 3 dB bandwidth, and a selectable continuous time linear equalizer (CTLE) to measure eye height and width. The pattern used for output eye diagram measurements is PRBS13Q. Unless specified otherwise the probabilities are relative to the number of PAM4 symbols measured. The following procedure should be used to obtain the eye height and eye width parameters, as illustrated by Figure 120E-13:

- 1) Capture the PRBS13Q using a clock recovery unit with a corner frequency of 4 MHz and slope of 20 dB/decade. The capture includes a minimum of 3 samples per symbol, or equivalent. Collect sufficient samples equivalent to at least 1.2 million PAM4 symbols to allow for construction of a normalized cumulative distribution function (CDF) to a probability of  $10^{-5}$  without extrapolation.
- 2) Apply the reference receiver including the appropriate CTLE to the captured signal. For module compliance, any single CTLE setting as described in 120E.3.2.1.1 that meets both eye width and eye height requirements is acceptable. For host compliance, any single CTLE setting as described in 120E.3.1.7 that meets both eye width and eye height requirements is acceptable.
- 3) Use the differential equalized signal from step 2) to construct the CDF of the signal threshold crossing for both the left edge (MID0CDFL) and right edge (MID0CDFR) of the middle eye, as a function of the distance from the center of the eye. Calculate the time center of the middle eye width (TCmid) as the mid-point in time between MID0CDFR and MID0CDFL with a value of  $10^{-3}$ . MID0CDFL and MID0CDFR are calculated as the cumulative sum of histograms of the zero crossing samples at the left and right edges of the middle eye normalized by the total number of sampled symbols. The MID0CDFL and MID0CDFR are equivalent to bath tub curves where the BER is plotted versus sampling time.
- 4) Use the differential equalized signal from step 2) to construct the CDF of the signal voltage of the middle eye within 0.025 UI of time TCmid, for both +1/3 (MIDCDF1) and -1/3 (MIDCDF0), as a function of the distance from the center of the eye. Calculate the eye height (Vmid) as the difference in voltage between MIDCDF1 and MIDCDF0 with a value of  $10^{-5}$ . Calculate the voltage center (VCmid) of the middle eye as the mid-point in voltage between MIDCDF1 and MIDCDF0 with a value of  $10^{-5}$ . MIDCDF0 and MIDCDF1 are calculated as the cumulative sum of histograms of the voltage at the top and bottom of the eye normalized by the total number of sampled bits.
- 5) Using the method in step 4), construct CDFs of signal voltages of the upper eye within 0.025 UI of time TCmid, and calculate the  $10^{-5}$  upper eye height (Vupp). Calculate the voltage center (VCupp) of the upper eye as the mid-point in voltage between UPPCDF1 and UPPCDF0 with a value of  $10^{-5}$ .
- 6) Using the method in step 4), construct CDFs of signal voltages of the lower eye within 0.025 UI of time TCmid, and calculate the  $10^{-5}$  lower eye height (Vlow). Calculate the voltage center (VClow) of the lower eye as the mid-point in voltage between LOWCDF1 and LOWCDF0 with a value of  $10^{-5}$ .
- 7) The eye height is defined as the minimum of Vmid, Vupp, and Vlow.

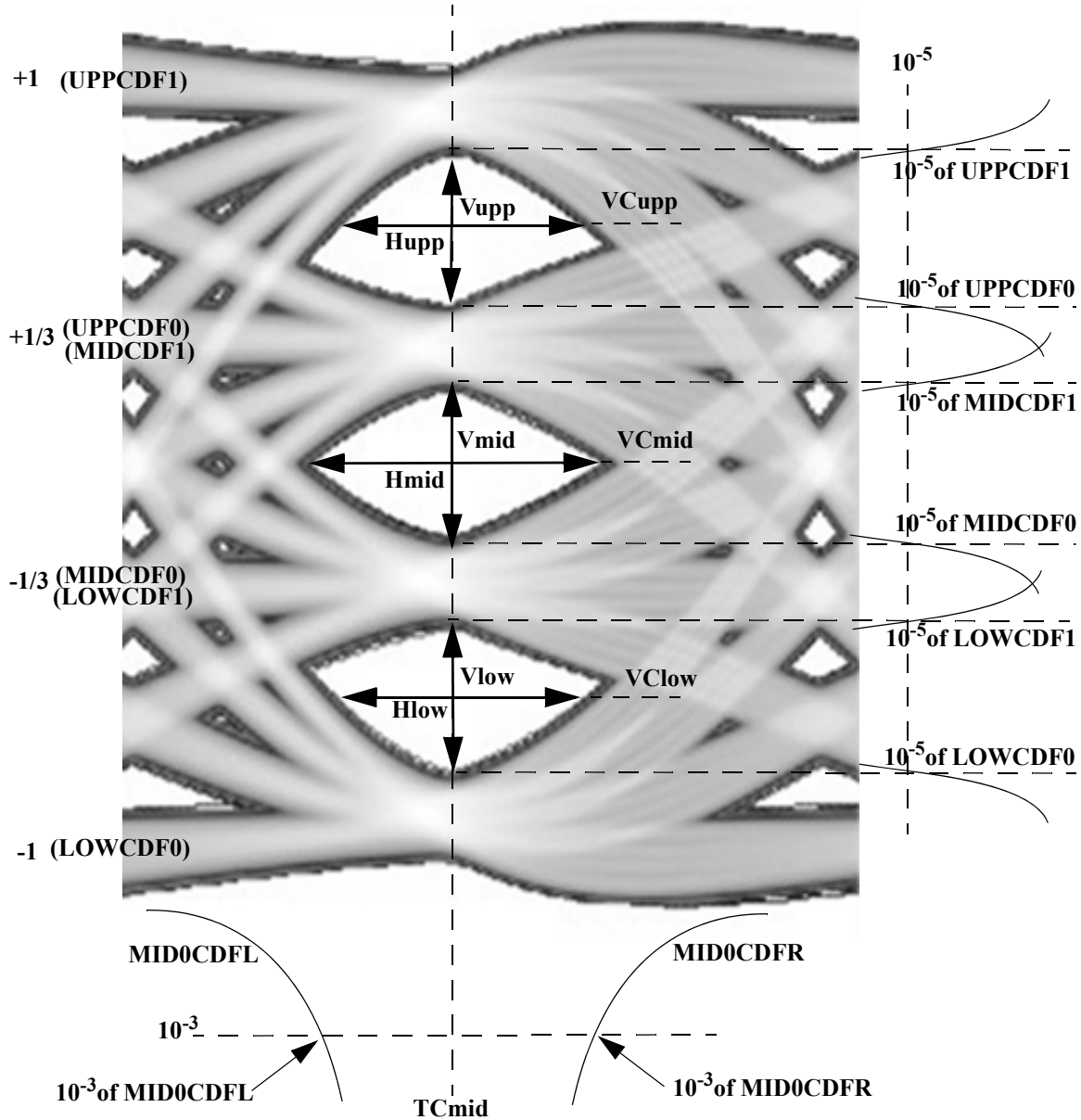


Figure 120E-13—PAM4 eye measurements

- 8) Use the differential equalized signal from step 2) to construct new CDFs of the signal for both the left edge (MIDCDFL) and right edge (MIDCDRL) of the middle eye at  $V_{Cmid}$ , as a function of the distance from the center of the eye. Calculate the middle eye width ( $H_{mid}$ ) as the difference in time between MIDCDRL and MIDCDFL with a value of  $10^{-5}$ .
- 9) Using the method in step 8), calculate the  $10^{-5}$  upper eye width ( $H_{upp}$ ) sliced at  $V_{Cupp}$ .
- 10) Using the method in step 8), calculate the  $10^{-5}$  lower eye width ( $H_{low}$ ) sliced at  $V_{Clow}$ .
- 11) The eye width is defined as the minimum of  $H_{mid}$ ,  $H_{upp}$ , and  $H_{low}$ .
- 12) Check the upper and lower eye alignment to the middle eye. Apply an ESMW wide mask centered on  $TC_{mid}$ , as illustrated by Figure 120E-14. The  $10^{-5}$  horizontal openings of the middle eye at  $V_{Cmid}$ , the upper eye at  $V_{Cupp}$ , and of the lower eye at  $V_{Clow}$  all have to extend beyond this mask.

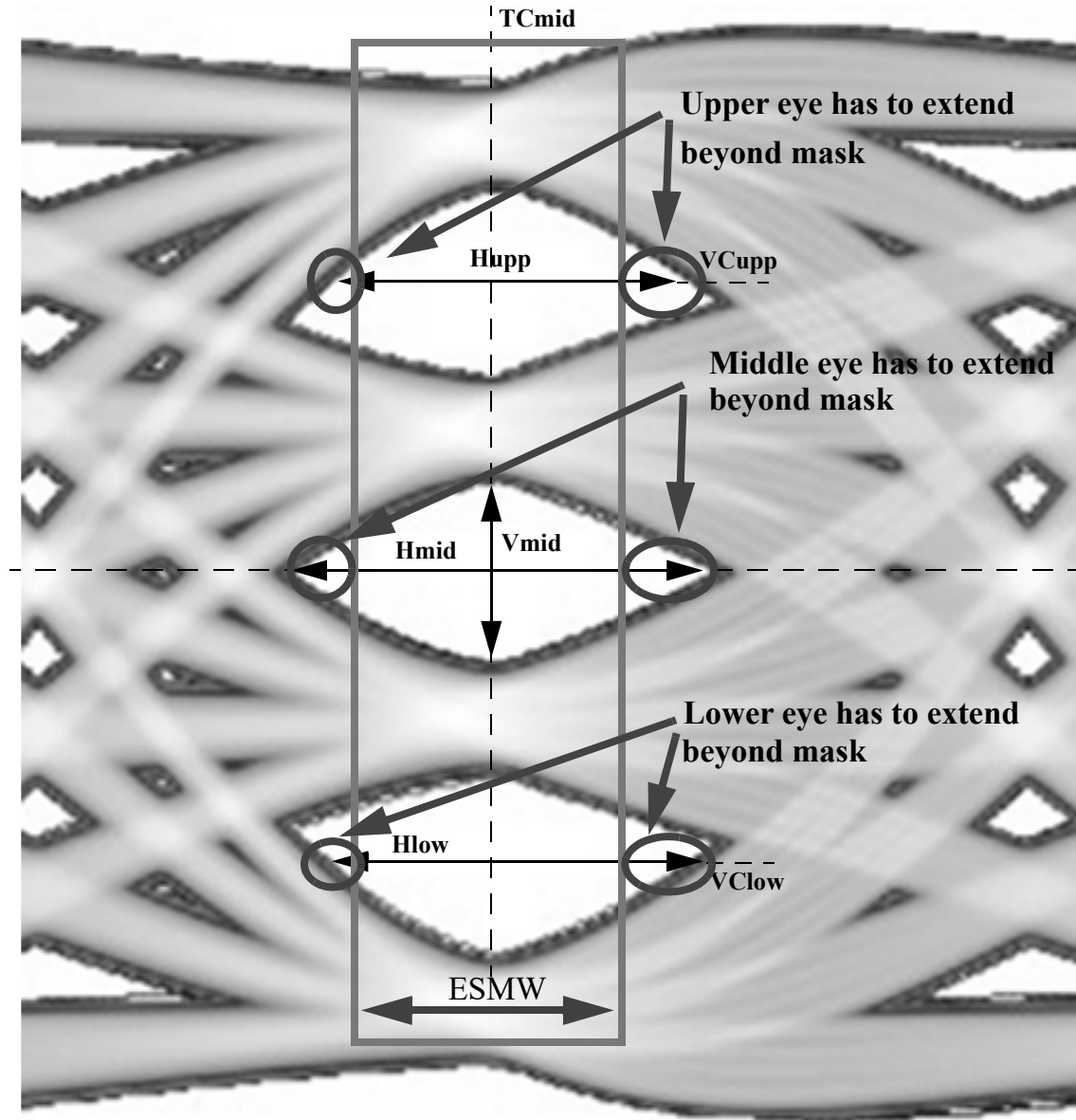


Figure 120E-14—PAM4 upper, middle, and lower eye mask

### 120E.4.3 Vertical eye closure

Vertical eye closure is defined by Equation (120E-4) for a PRBS13Q differential equalized signal captured and processed according to 120E.4.2.

$$VEC = 20\log_{10}\left(\max\left(\frac{AV_{upp}}{V_{upp}}, \frac{AV_{mid}}{V_{mid}}, \frac{AV_{low}}{V_{low}}\right)\right) \text{ (dB)} \quad (120E-4)$$

where

$VEC$  is the vertical eye closure, in dB

|                   |                                                                                                                                       |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| $V_{\text{upp}}$  | is the $10^{-5}$ upper eye height per 120E.4.2                                                                                        |
| $V_{\text{mid}}$  | is the $10^{-5}$ middle eye height per 120E.4.2                                                                                       |
| $V_{\text{low}}$  | is the $10^{-5}$ lower eye height per 120E.4.2                                                                                        |
| $AV_{\text{upp}}$ | is the amplitude of the upper eye ( $AV_{\text{upp}}$ ), equal to $VM3-VM2$                                                           |
| $AV_{\text{mid}}$ | is the amplitude of the middle eye ( $AV_{\text{mid}}$ ), equal to $VM2-VM1$                                                          |
| $AV_{\text{low}}$ | is the amplitude of the lower eye ( $AV_{\text{low}}$ ), equal to $VM1-VM0$                                                           |
| $VM3$             | is the mean of the differential equalized signal above $VC_{\text{upp}}$ within 0.025 UI of $TC_{\text{mid}}$                         |
| $VM2$             | is the mean of the differential equalized signal between $VC_{\text{upp}}$ and $VC_{\text{mid}}$ within 0.025 UI of $TC_{\text{mid}}$ |
| $VM1$             | is the mean of the differential equalized signal between $VC_{\text{mid}}$ and $VC_{\text{low}}$ within 0.025 UI of $TC_{\text{mid}}$ |
| $VM0$             | is the mean of the differential equalized signal below $VC_{\text{low}}$ within 0.025 UI of $TC_{\text{mid}}$                         |
| $VC_{\text{upp}}$ | is the voltage center of the upper eye per 120E.4.2                                                                                   |
| $VC_{\text{mid}}$ | is the voltage center of the middle eye per 120E.4.2                                                                                  |
| $VC_{\text{low}}$ | is the voltage center of the lower eye per 120E.4.2                                                                                   |
| $TC_{\text{mid}}$ | is the time center of the middle eye width per 120E.4.2                                                                               |

## 120E.5 Protocol implementation conformance statement (PICS) proforma for Annex 120E, Chip-to-module 200 Gb/s four-lane Attachment Unit Interface (200GAUI-4 C2M) and 400 Gb/s eight-lane Attachment Unit Interface (400GAUI-8 C2M)<sup>309</sup>

### 120E.5.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Annex 120E, Chip-to-module 200 Gb/s four-lane Attachment Unit Interface (200GAUI-4 C2M) and 400 Gb/s eight-lane Attachment Unit Interface (400GAUI-8 C2M), shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

### 120E.5.2 Identification

#### 120E.5.2.1 Implementation identification

|                                                                                                                                                                                                                                                                                                         |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Supplier <sup>1</sup>                                                                                                                                                                                                                                                                                   |  |
| Contact point for inquiries about the PICS <sup>1</sup>                                                                                                                                                                                                                                                 |  |
| Implementation Name(s) and Version(s) <sup>1,3</sup>                                                                                                                                                                                                                                                    |  |
| Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) <sup>2</sup>                                                                                                                                                     |  |
| <p>NOTE 1—Required for all implementations.</p> <p>NOTE 2—May be completed as appropriate in meeting the requirements for the identification.</p> <p>NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).</p> |  |

#### 120E.5.2.2 Protocol summary

|                                                                                                                                                                                                                       |                                                                                                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Identification of protocol standard                                                                                                                                                                                   | IEEE Std 802.3-2022, Annex 120E, Chip-to-module 200 Gb/s four-lane Attachment Unit Interface (200GAUI-4 C2M) and 400 Gb/s eight-lane Attachment Unit Interface (400GAUI-8 C2M) |
| Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS                                                                                                       |                                                                                                                                                                                |
| <p>Have any Exception items been required? No <input type="checkbox"/> Yes <input type="checkbox"/></p> <p>(See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3-2022.)</p> |                                                                                                                                                                                |

|                   |  |
|-------------------|--|
| Date of Statement |  |
|-------------------|--|

<sup>309</sup>Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

### 120E.5.3 Major capabilities/options

| Item | Feature                                 | Subclause | Value/Comment                                                                                                                        | Status | Support |
|------|-----------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------|--------|---------|
| NOL  | Number of differential AC-coupled lanes | 120E.1    | Four independent data paths in each direction for 200GAUI-4 C2M and eight independent data paths in each direction for 400GAUI-8 C2M | M      | Yes [ ] |
| BER  | Meets BER requirement                   | 120E.1.1  | See 120E.1.1                                                                                                                         | M      | Yes [ ] |
| ADE  | Adaptive Equalizer                      | 120E.3.4  | See 120E.3.4                                                                                                                         | M      | Yes [ ] |

### 120E.5.4 PICS proforma tables for Chip-to-module 200 Gb/s four-lane Attachment Unit Interface (200GAUI-4 C2M) and 400 Gb/s eight-lane Attachment Unit Interface (400GAUI-8 C2M)

#### 120E.5.4.1 Host output

| Item | Feature                                                        | Subclause  | Value/Comment                                                                                         | Status | Support |
|------|----------------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------|--------|---------|
| TH1  | Signaling rate                                                 | 120E.3.1.1 | 26.5625 GBd $\pm$ 100 ppm per lane                                                                    | M      | Yes [ ] |
| TH2  | DC common-mode output voltage                                  | 120E.3.1   | Between $-0.3$ V and $2.8$ V with respect to signal ground                                            | M      | Yes [ ] |
| TH3  | Peak-to-peak differential output voltage, transmitter disabled | 120E.3.1.2 | Less than or equal to $35$ mV                                                                         | M      | Yes [ ] |
| TH4  | Peak-to-peak differential output voltage                       | 120E.3.1.2 | $880$ mV (max)                                                                                        | M      | Yes [ ] |
| TH5  | AC common-mode output voltage                                  | 120E.3.1   | $17.5$ mV RMS with respect to signal ground                                                           | M      | Yes [ ] |
| TH6  | Eye width, all 3 eyes                                          | 120E.3.1   | $0.22$ UI                                                                                             | M      | Yes [ ] |
| TH7  | Eye height, all 3 eyes                                         | 120E.3.1   | $32$ mV                                                                                               | M      | Yes [ ] |
| TH8  | Vertical eye closure                                           | 120E.3.1   | $12$ dB                                                                                               | M      | Yes [ ] |
| TH9  | Differential output return loss                                | 120E.3.1.3 | Meets Equation (83E–2) constraints                                                                    | M      | Yes [ ] |
| TH10 | Common to differential mode conversion                         | 120E.3.1.3 | Meets Equation (83E–3) constraints                                                                    | M      | Yes [ ] |
| TH11 | Differential termination mismatch                              | 120E.3.1   | Less than $10\%$                                                                                      | M      | Yes [ ] |
| TH12 | Transition time                                                | 120E.3.1.5 | Greater than or equal to $10$ ps                                                                      | M      | Yes [ ] |
| TH13 | Reference impedance for output return loss                     | 120E.3.1.3 | $100\ \Omega$ .                                                                                       | M      | Yes [ ] |
| TH14 | Crosstalk source                                               | 120E.3.1.6 | Asynchronous crosstalk source using PRBS13Q or PRBS31Q pattern, or valid 200GBASE-R/400GBASE-R signal | M      | Yes [ ] |

#### 120E.5.4.2 Module output

| Item | Feature                                    | Subclause    | Value/Comment                                                                                         | Status | Support |
|------|--------------------------------------------|--------------|-------------------------------------------------------------------------------------------------------|--------|---------|
| TM1  | Signal rate                                | 120E.3.1.1   | 26.5625 GBd $\pm$ 100 ppm per lane                                                                    | M      | Yes [ ] |
| TM2  | AC common-mode output voltage              | 120E.3.1     | 17.5 mV RMS with respect to signal ground                                                             | M      | Yes [ ] |
| TM3  | Peak-to-peak differential output voltage   | 120E.3.1.2   | 900 mV (max)                                                                                          | M      | Yes [ ] |
| TM4  | Near-end eye width, all 3 eyes             | 120E.3.2.1   | 0.265 UI                                                                                              | M      | Yes [ ] |
| TM5  | Near-end eye height, all 3 eyes            | 120E.3.2.1   | 70 mV                                                                                                 | M      | Yes [ ] |
| TM6  | Far-end eye width, all 3 eyes              | 120E.3.2.1   | 0.2 UI                                                                                                | M      | Yes [ ] |
| TM7  | Far-end eye height, all 3 eyes             | 120E.3.2.1   | 30 mV                                                                                                 | M      | Yes [ ] |
| TM8  | Far-end pre-cursor ISI ratio               | 120E.3.2.1.2 | −4.5% to +2.5%                                                                                        | M      | Yes [ ] |
| TM9  | Differential output return loss            | 120E.3.1.3   | Meets Equation (83E–2) constraints                                                                    | M      | Yes [ ] |
| TM10 | Common to differential mode conversion     | 120E.3.1.3   | Meets Equation (83E–3) constraints                                                                    | M      | Yes [ ] |
| TM11 | Differential termination mismatch          | 120E.3.1     | Less than 10%                                                                                         | M      | Yes [ ] |
| TM12 | Transition time                            | 120E.3.1.5   | Greater than or equal to 9.5 ps                                                                       | M      | Yes [ ] |
| TM13 | Reference impedance for output return loss | 120E.3.1.3   | 100 $\Omega$ .                                                                                        | M      | Yes [ ] |
| TM14 | Crosstalk source                           | 120E.3.2.1   | Asynchronous crosstalk source using PRBS13Q or PRBS31Q pattern, or valid 200GBASE-R/400GBASE-R signal | M      | Yes [ ] |

#### 120E.5.4.3 Host input

| Item | Feature                    | Subclause | Value/Comment | Status | Support |
|------|----------------------------|-----------|---------------|--------|---------|
| RH1  | Host input characteristics | 120E.3.3  | Table 120E–4  | M      | Yes [ ] |

#### 120E.5.4.4 Module input

| Item | Feature                      | Subclause | Value/Comment | Status | Support |
|------|------------------------------|-----------|---------------|--------|---------|
| RM1  | Module input characteristics | 120E.3.4  | Table 120E–7  | M      | Yes [ ] |